RAJALAKSHMI ENGINEERING COLLEGE

Akshaya S 251001013-2025-IT

A2

Started on	Tuesday, 25 August 2026, 11:10 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 11:13 AM
Time taken	3 mins 14 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n, rev, remainder, count = 0;

    scanf("%d", &n);

    rev = 0;
    count++;              // rev = 0

    while (n != 0)
    {
        count++;          // while condition

        remainder = n % 10;
        count++;

        rev = rev * 10 + remainder;
        count++;

        n /= 10;
        count++;
    }

    count++;              // final while condition
    count++;              // original print(rev)

    printf("%d", count);  // counter printf not counted

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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